Deepika V

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.17
KENDRIYA VIDYALAYA NO. 2 UPPAL HYDERABAD TL	Class XII (CBSE)	2020	83.00%
DEFENCE LAB SCHOOL KANCHANBAGH HYDERABAD TL	Class X (CBSE)	2018	86.80%

EXPERIENCE

PRICE WATCH.AI | Associate, Commodity Pricing

- Analyzed **commodity datasets and pricing movements** across Feedstock and Polymer industries using AI-driven approaches to develop pricing reports supporting sourcing and procurement decisions

BEROE INC | Associate, Commodity Pricing

- Analyzed **supply, cost structures and pricing trends** of feedstock-derived products in Feedstock and Polymer industries to support procurement decisions

INTERNSHIP

VEEKESY FOOTWEAR -- Process Engineering Intern

· Analyzed **process and material factors driving rejects** in the EVA production unit, studying reject types and data patterns to identify sources of process variation; supported material improvement through EVA blend preparation.

FACT -- Manufacturing Process Intern

- Gained exposure to fertilizer and chemical **process operations, PFDs, process equipment and instrumentation** across ammonium sulphate, sulphuric acid and ammonia production units; reviewed **energy conservation and pollution control** practices.

PROJECTS

M.Tech Thesis |Physics-Based Modelling of Biomolecular Condensates

- Developing a **Python-based image analysis pipeline** using computer vision for image processing, segmentation and quantitative feature extraction.

- Integrating **image-derived data with physics-based models** using FEniCS, thermodynamics, reaction engineering and transport phenomena for **biophysical parameter estimation**.

B.Tech Thesis|3D Electrochemical Reactor for Heavy Metal Removal

- Designed and fabricated a **3D electrochemical reactor**; performed **Linear Sweep Voltammetry** for kinetic parameters and **design of experiments** across flow rates and current levels to evaluate lead removal.

Statistical Analysis & Design of Experiments for Process Optimization

- Applied **DOE and Response Surface Methodology** with regression, **ANOVA**, residual, main-effect and interaction analysis to identify significant process factors and factor combinations meeting process specifications.

Artificial Neural Network-Based Gas Turbine Emission Prediction | GitHub

- Analyzed operating data in **Python** using EDA, correlation analysis, data cleaning, outlier analysis and feature scaling and developed a **multi-output ANN** (TensorFlow/Keras) evaluated on MSE and R^2 to study CO and NOX emissions.

CFD Simulation -- Laminar Natural Convection

- Modelled buoyancy-driven natural convection in a 2D refrigerator cavity using **ANSYS Fluent** (Boussinesq approximation, SIMPLE coupling), analyzing velocity, temperature and pressure fields with and without internal shelves.

Monte Carlo Simulation of the 2D Ising Model | **GitHub**

- Developed a **Python**-based **Monte Carlo simulation** (Glauber/Metropolis dynamics, periodic boundary conditions) to study effect of interaction strength, field and lattice size on magnetization, susceptibility and specific heat.

Process Design & Techno-Economic Analysis of Bisphenol A Production

- Developed an **Aspen Plus PFD** and performed **material and energy balances**; evaluated **process integration, recycle and heat integration** for process efficiency, along with **techno-economic analysis** of capital and utility costs.

SKILLS

Data Analysis & Statistics: Python | NumPy | Pandas | Scikit-learn | Matplotlib | Minitab | Regression Analysis | ANOVA | Design of Experiments | ANN | TensorFlow/Keras | NSGA-II

Simulation & Tools: ANSYS Fluent | Aspen Plus | Aspen HYSYS | MATLAB | FEniCS | AutoCAD | Aftermath | MS Excel

RELEVANT COURSES

Process Control | Transport Phenomena | Molecular Thermodynamics | Heat & Mass Transfer | Process Design & Simulation | Statistical Analysis & Design of Experiments | Numerical Methods | Machine Learning | Computational Methods | Process Safety

SCHOLASTIC ACHIVEMENTS

- **Samsung EnnovateX AX Hackathon 2026:** RL-based reasoning pipeline (GRPO/PPO) for small language models
- **Competitions:** Amazon ML Summer School Challenge | NEO Olympiad
- **Certifications:** Six Sigma Yellow Belt | Aspen Plus V11 | MATLAB Onramp | Heat Exchanger Design & Inspection

POSITION OF RESPONSIBILITY

- **Career & Placement Coordinator** | Office of Career Services | IIT Hyderabad
- **Operations Coordinator** | ITECH/ITC Incubator | IIT Hyderabad

EXTRA CURRICULARS AND HOBBIES

Extra-curricular : Silver Medal -- DIeSTA Volleyball Tournament | Bronze Medal -- 1500 m Run, District Level | NCC | IIChE

Hobbies: Travelling | Dancing | Doodling | Crochet